

ECONOMICS

Week 03

Interest Calculations

Sardar Shahbaz A. Naqvi

The Origins of Interest

- Like taxes, interest has existed from earliest recorded human history.
- Records reveal its existence in Babylon in 2000 b.c
- During the Middle Ages, interest taking on loans of money was generally outlawed on scriptural grounds.
- In 1536, the Protestant theory of usury was established by John Calvin, and it refuted the notion that interest was unlawful.
- Consequently, interest taking again became viewed as an essential and legal part of doing business

Simple Interest

When the total interest earned or charged is linearly proportional to the initial amount of the loan (principal), the interest rate, and the number of interest periods for which the principal is committed, the interest and interest rate are said to be simple.

Simple Interest

When simple interest is applicable, the total interest I , earned or paid may be computed using the formula

$$I = (P)(N)(i)$$

Where,

P = principal amount lent or borrowed;

N = number of interest periods (e.g., years);

i = interest rate per interest period.

Simple Interest

The total amount repaid at the end of N interest periods is $P + I$. Thus, if \$1,000 were loaned for three years at a simple interest rate of 10% per year, the interest earned would be

$$I = \$1,000 \times 3 \times 0.10 = \$300$$

The total amount owed at the end of three years would be $\$1,000 + \$300 = \$1,300$.

Simple Interest

$$I = n \times i \times P$$

I = Total amount of simple Interest

n = life of the loan

i = interest rate (expressed as percentage or decimal)

P = Principal amount

$$F = P + I$$

$$F = P + n i P$$

$$F = P (1 + ni)$$

F = Future Value

Simple Interest - Problems

1. The ABC Company deposited \$100,000 in a bank account on June 15 and withdrew a total of \$115,000 exactly one year later. Compute: a) the interest rate which the ABC company received from the \$100,000 investment & b) annual interest which the ABC Company was paid.
2. What is the annual rate of simple interest if \$265 is earned in four months on an investment of \$15,000?
3. Determine the principle that would have to be invested to provide \$200 of simple interest income at the end of two years if the annual interest rate is 9%.
4. At what annual interest rate is \$500 one year ago equivalent to \$600 today?

Problem 1

The ABC Company deposited \$100,000 in a bank account on June 15 and withdrew a total of \$115,000 exactly one year later. Compute:

- a) The interest rate which the ABC company received from the \$100,000 investment
- b) Annual interest which the ABC Company was paid.

Data:

$P = \$100,000$

$F_1 = \$115,000$

$n = 1$ year

Solution

$i = ?$

$I = ?$

$F = P + I$

$$115,000 = 100,000 (1 + i (1))$$

$$i = 15\%$$

$$I = \$15,000$$

Formula

$$F = P + I$$

$$F = P + ni P$$

$$F = P (1 + ni)$$

Problem 2

What is the annual rate of simple interest if \$265 is earned in four months on an investment of \$15,000?

Data:

$I = \$265$ after 4 months

$P = \$15,000$

$n = 4/12 = 0.33$

$i = ?$

Answer

$I = P \times i \times n$

$i = I / (P \times n)$

$i = 5.3\%$ annual rate

Formula

$$F = P + I$$

$$F = P + niP$$

$$F = P(1 + ni)$$

Problem 3

Determine the principle that would have to be invested to provide \$200 of simple interest income at the end of two years if the annual interest rate is 9%.

Data:

- $I = \$200$ after 2 years
- $n = 2$ years
- $i = 9\%$
- $P = ?$

Solution

- $I_2 = Pin$
- $200 = P \times 0.09 \times 2$
- $P = \$1111$

Formula

$$\begin{aligned} F &= P + I \\ F &= P + niP \\ F &= P(1 + ni) \end{aligned}$$

Problem 4

At what annual interest rate is \$500 one year ago equivalent to \$600 today?

Data:

- $P = \$500$
- $n = 1$ year
- $F = \$600$
- $i = ?$

Solution

$$I = F - P$$

$$I = Pin$$

$$i = 0.2 = 20\%$$

Formula

$$F = P + I$$

$$F = P + niP$$

$$F = P(1 + ni)$$

Compound Interest

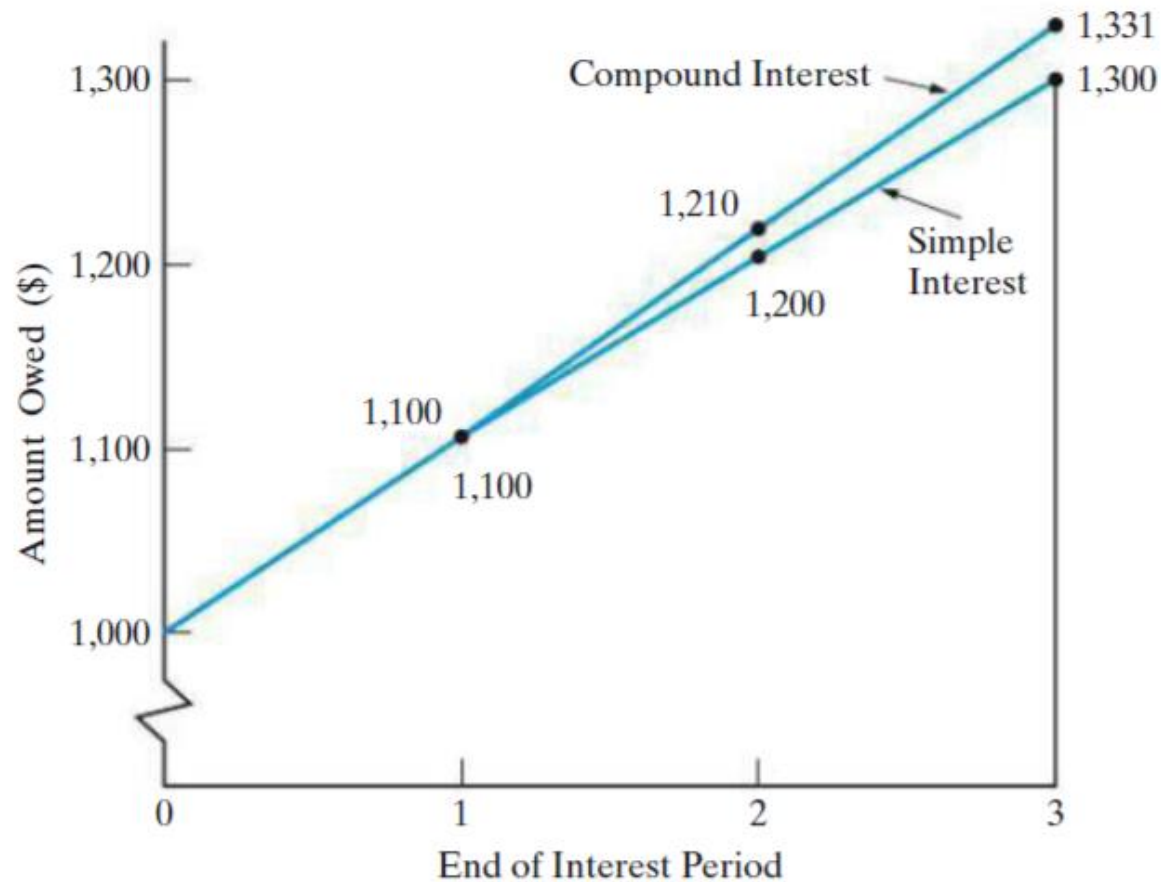
Whenever the interest charge for any interest period (a year, for example) is based on the remaining principal amount plus any accumulated interest charges up to the beginning of that period, the interest is said to be compound.

Compound Interest

\$1,000 loaned for three periods at an interest rate of 10% compounded each period:

Period	(1) Amount Owed at Beginning of Period	(2) = (1) × 10% Interest Amount for Period	(3) = (1) + (2) Amount Owed at End of Period
1	\$1,000	\$100	\$1,100
2	\$1,100	\$110	\$1,210
3	\$1,210	\$121	\$1,331

Illustration of Simple versus Compound Interest



Compound Interest

Year 1

$$I_1 = niP$$

$$F_1 = P + I_1$$

$$F_1 = P + niP \text{ let } n = 1$$

$$F_1 = P(1+i)$$

Year 2

$$I_2 = iF_1$$

$$I_2 = iP(1+i)$$

$$F_2 = F_1 + I_2$$

$$F_2 = P + iP + iP(1+i)$$

$$F_2 = P(1+i)^2$$

Compound Interest

Year 3

$$I_3 = i F_2$$

$$I_3 = i P (1+i)^2$$

$$F_3 = F_2 + I_3$$

$$F_3 = P (1+i)^2 + i P (1+i)^2$$

$$F_3 = P (1+i)^3$$

$$F_n = P (1+i)^n$$

Problems:

1. How long would it take for an investor to double his money at 10% interest per year, compounded annually?
2. Suppose that a man lends \$1000 for four years at 12% per year simple interest. At end of four years, he invests the entire amount which he then has for 10 years at 8% interest per year, compounded annually. How much money will he have at the end of the 14-year period?
3. Supposed that the interest rate is 10% per year, compounded annually. What is the minimum amount of money that would have to be invested for a two-year period in order to earn \$300 in interest?

Problem 1

How long would it take for an investor to double his money at 10% interest per year, compounded annually?

Data:

- $F = 2P$
- $i = 0.1$
- $n = ?$
- $F_n = P (1+0.1)^n$
- $2P = P (1.1)^n$
- $n = \ln 2 / \ln 1.1 = 7.23 \text{ years}$

Problem 2

Suppose that a man lends \$1000 for four years at 12% per year simple interest. At end of four years, he invests the entire amount which he then has for 10 years at 8% interest per year, compounded annually. How much money will he have at the end of the 14-year period?

- **DATA**

- $P = \$1000$
- $n = 4$ years
- $i = 12\%$ simple Interest=
- $F_4 = P (1+in)$
- $F_4 = 1000 (1+0.12 \times 4) = \1480 after four years

Formula

Reinvested F_4

- $n = 10$ years
- $i = 8\%$
- $F_{10} = 1480 (1 + .08)^{10}$
- $F_{10} = \$ 3195.21$

$$F = P + I$$

$$F = P + ni P$$

$$F = P (1+ni)$$

$$F_n = P (1+i)^n$$

Amount to get after 14 years is \$3195.21

Problem 3

Supposed that the interest rate is 10% per year, compounded annually. What is the minimum amount of money that would have to be invested for a two-year period in order to earn \$300 in interest?

DATA

- $P = ?$
- $n = 2$ years
- $i = 10\%$
- $I_2 = \$300$

Solution

$$I_2 = i \times F_1$$

$$F_1 = P(1+i)^1$$

$$I_2 = i \times P(1+i)^1$$

$$P = \$ 2727.27$$

Formula

$$F_n = P (1+i)^n$$

$$F_1 = P + I_1$$

$$F_1 = P + ni P \text{ let } n = 1$$

$$F_1 = P (1+i)$$

$$I_2 = i F_1$$

$$I_2 = i P (1+i)$$

$$F_2 = F_1 + I_2$$

$$F_2 = P + i P + i P (1+i)$$

$$F_2 = P (1+i)^2$$

Interest Rate Usage

Time Value of Money

- $(\text{Future Value})_n = \text{Initial Value} (1 + \text{rate of money increase})^n$

Inflation

- $FC = PC (1 + \tilde{\lambda})^n$
 - $\tilde{\lambda}$ – Inflation Rate
 - FC – Future Cost
 - PC – Present Cost